Measure Theoretic Probability
Lesson 14
Random Variables and Measurability

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Outline

1. Three Levels of Random Variables
2. E.g., Three Coin Flips
3. Simple Random Variables
4. Indicator Random Variables
5. Measurability with Respect to a Sub-σ-Field
6. Summary and Exercises

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Three Definitions of a Random Variable

Level 1 (beginner)

A variable taking different values with different probabilities. For example, $X = 1$ with probability $\frac{1}{4}$, $X = 2$ with probability $\frac{1}{3}$, $X = 3$ with probability $\frac{5}{12}$.

Level 2 (intermediate)

A **function** from the outcomes of an experiment to numbers. Flip a coin once, $\Omega = \{H, T\}$, and let

$$X(H) = 1, \qquad X(T) = 0.$$

Level 3 (measure theoretic)

Given a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, a **random variable** is a function

$$X : \Omega \to \mathbb{R} \quad \text{that is } \textbf{measurable} \text{ with respect to } \mathcal{F}.$$

Today's task: say precisely what "measurable" means.

By Convention, R.V are written with capital letters (X, Y, Z);lowercase denote R

What "Measurable" Should Mean

Recall why we have σ-fields

The sets in $\mathcal{F}$ are the events whose probabilities we know (or can derive from what we know by complements, countable additivity, and so on).

Informal idea

X is **measurable with respect to** $\mathcal{F}$ ("$\mathcal{F}$-measurable", or just "measurable") if every probability we want to compute for X can be obtained from a set that lies in $\mathcal{F}$.

Remark

Measurability is always relative to a σ-field. The same function can be measurable for one σ-field and not for another, as the next example shows.

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Limited Information About a Coin

The experiment

A coin is flipped three times; $\Omega = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$. We know nothing about the coin, but someone tells us $\mathbb{P}(A)$ for

$$A = \{\text{at least two heads}\} = \{HHH, HHT, HTH, THH\}.$$

What we can then derive

$\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$, $\mathbb{P}(\varnothing) = 0$, $\mathbb{P}(\Omega) = 1$. So the sets we can measure form

$$\mathcal{F}_1 = \{\varnothing,\ A,\ A^c,\ \Omega\} = \sigma(A).$$

Question

Which random variables on Ω can we study using only $\mathcal{F}_1$?

Two Random Variables, Two Outcomes

$X = 1$ if at least two heads, 0 otherwise

- $\mathbb{P}(X = 1) = \mathbb{P}(A)$, which is known.
- $\mathbb{P}(X = 0) = \mathbb{P}(A^c) = 1 - \mathbb{P}(A)$, which is known.

All probabilities for X come from sets in $\mathcal{F}_1$, so X is measurable with respect to $\mathcal{F}_1$: **a valid random variable on** $(\Omega, \mathcal{F}_1, \mathbb{P})$.

X = number of heads in three flips (values $0, 1, 2, 3$)

$\mathbb{P}(X = 1)$ requires the set $\{HTT, THT, TTH\}$, but it is not in $\mathcal{F}_1$: its probability was never given to us and cannot be derived. Likewise $\{X = 3\} = \{HHH\}$ is a singleton outside $\mathcal{F}_1$.

Conclusion

The number of heads is **not** $\mathcal{F}_1$-measurable, hence not a random variable on $(\Omega, \mathcal{F}_1, \mathbb{P})$.

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Definition: Simple Random Variable

Definition

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A function $X : \Omega \to \mathbb{R}$ is a **simple random variable** if

1. X takes only **finitely many** values (not merely countably many), and
2. for every $x \in \mathbb{R}$,
$$\{\omega \in \Omega : \ X(\omega) = x\} \in \mathcal{F}.$$

Remarks

- The values need not be integers: X could take values $\pi,\ e,\ 7.2$.
- The set in (2) is the **inverse image** of x under X, written $X^{-1}(\{x\})$.
- It is abbreviated $\{X = x\}$: a **subset of** Ω, not a number.

Revisiting the Coin Example with Inverse Images

$X = 1$ if at least two heads, 0 otherwise, and $\mathcal{F}_1 = \{\varnothing, A, A^c, \Omega\}$

$$\{X = 1\} = A \in \mathcal{F}_1, \qquad \{X = 0\} = A^c \in \mathcal{F}_1, \qquad \{X = 7.2\} = \varnothing \in \mathcal{F}_1.$$

Every other x also gives $\varnothing$. So X is $\mathcal{F}_1$-measurable. No probabilities were needed: we only checked that sets lie in the σ-field.

$X =$ number of heads

- With $\mathcal{F}_1$: $\{X = 3\} = \{HHH\} \notin \mathcal{F}_1$, so X is **not** $\mathcal{F}_1$-measurable.
- With $\mathcal{F}_2 = \mathcal{P}(\Omega)$ (the power set, every subset of Ω): all inverse images are subsets of Ω, so they lie in $\mathcal{F}_2$. Hence X **is** $\mathcal{F}_2$-measurable.

A question for next time

Is there a **smallest** σ-field for which this X is measurable? Yes: the σ-field generated by X, the topic of the next lesson.

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Indicator Functions

Definition

For a set $A \subseteq \mathbb{R}$ (no σ-field involved yet), the **indicator function** is

$$I_A(x) = \begin{cases} 1, & x \in A, \\ 0, & x \notin A. \end{cases}$$

Notation

Other common notations: 1_A, χ_A, and the Iverson bracket $[x \in A]$. We use I_A.

Indicator random variable

On $(\Omega, \mathcal{F}, \mathbb{P})$, for any set $A \subseteq \Omega$ define $X(\omega) = I_A(\omega)$.

When Is an Indicator Measurable?

Claim

$X = I_A$ is $\mathcal{F}$-measurable **if and only if** $A \in \mathcal{F}$.

Proof

The inverse images of the possible values are

$$\{X = 1\} = A, \qquad \{X = 0\} = A^c, \qquad \{X = x\} = \varnothing \text{ for all other } x.$$

- ($\Rightarrow$) If X is measurable then $\{X = 1\} = A \in \mathcal{F}$.
- ($\Leftarrow$) If $A \in \mathcal{F}$ then $A^c \in \mathcal{F}$ (complements), and $\varnothing \in \mathcal{F}$ (every σ-field contains it). All three inverse images are in $\mathcal{F}$. □

Moral

Indicators are the building blocks of random variables: the event A is "observable" precisely when I_A is a random variable.

Sums of Indicators over a Partition

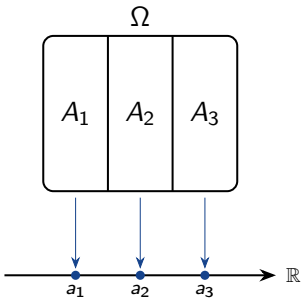
Construction

Let $A_1, \ldots, A_n \in \mathcal{F}$ be a **finite partition** of Ω (pairwise disjoint, with union Ω), and let $a_1, \ldots, a_n \in \mathbb{R}$ be distinct. Define

$$X(\omega) = \sum_{i=1}^{n} a_i \, I_{A_i}(\omega).$$

It is a simple random variable

- It takes the finite set of values $a_1, \ldots, a_n$.
- $\{X = a_i\} = A_i \in \mathcal{F}$, and $\{X = x\} = \varnothing \in \mathcal{F}$ if $x \notin \{a_1, \ldots, a_n\}$.



Every Simple Random Variable Is a Sum of Indicators

Representation

Let X be simple, with distinct values $a_1, \ldots, a_n$, and put

$$A_i = \{X = a_i\} \in \mathcal{F}.$$

The A_i are disjoint (each ω has only one value $X(\omega)$) and cover Ω, so they form a finite partition, and

$$X = \sum_{i=1}^{n} a_i \, I_{A_i}.$$

Why "distinct"?

If two terms share a value, say $a_1 = a_3$, merge $A_1 \cup A_3$ into one set with a single coefficient.

Example: number of heads, with $\mathcal{F} = \mathcal{P}(\Omega)$

$$X = 0 \cdot I_{\{TTT\}} + 1 \cdot I_{\{HTT,THT,TTH\}} + 2 \cdot I_{\{HHT,HTH,THH\}} + 3 \cdot I_{\{HHH\}}.$$

For $\omega = HHT$ only the third indicator equals 1, so $X(HHT) = 2$.

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Sub-σ-Fields

Definition

Let X be a simple random variable on $(\Omega, \mathcal{F}, \mathbb{P})$ and let $\mathcal{G} \subseteq \mathcal{F}$ be a σ-field (a **sub-σ-field**). X is **$\mathcal{G}$-measurable** if

$$\{X = x\} \in \mathcal{G} \qquad \text{for every } x \in \mathbb{R}.$$

Remark

Since $\mathcal{G} \subseteq \mathcal{F}$, this is stronger than $\mathcal{F}$-measurability: the inverse images must lie in the smaller collection $\mathcal{G}$.

Example

In the coin example, the number of heads is $\mathcal{F}_2$-measurable but not $\mathcal{F}_1$-measurable. The σ-field must be big enough to contain the inverse images.

Inverse Images of Arbitrary Subsets of $\mathbb{R}$

Proposition

Let X be simple and $\mathcal{G}$-measurable. For **any** $H \subseteq \mathbb{R}$ (not necessarily countable or Borel),

$$\{X \in H\} = \{\omega : \ X(\omega) \in H\} \in \mathcal{G}.$$

Proof

Write

$$\{X \in H\} = \bigcup_{x \in H} \{X = x\}.$$

This union may look uncountable, but X takes only finitely many values, say $a_1, \ldots, a_n$, and $\{X = x\} = \varnothing$ for every other x. So

$$\{X \in H\} = \bigcup_{i: \ a_i \in H} \{X = a_i\},$$

a **finite** union of sets in $\mathcal{G}$. A σ-field is closed under finite unions (a special case of countable unions, padding with $\varnothing$), so $\{X \in H\} \in \mathcal{G}$. □

Why this matters

For simple X, checking the single values $\{X = x\}$ is enough to control **every** event of the form $\{X \in H\}$.

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Summary

Key results

Concept / Result	Content
Random variable	$X : \Omega \to \mathbb{R}$ measurable with respect to $\mathcal{F}$
Simple random variable	finitely many values and $\{X = x\} \in \mathcal{F}$ for all $x \in \mathbb{R}$
Inverse image	$\{X = x\} = \{\omega : \ X(\omega) = x\} \subseteq \Omega$
Indicator	I_A is measurable $\iff A \in \mathcal{F}$
Representation	simple $X = \sum_i a_i I_{A_i}$ with $A_i = \{X = a_i\}$ a finite partition
Sub-σ-field	X is $\mathcal{G}$-measurable if $\{X = x\} \in \mathcal{G}$ for all x
Arbitrary $H \subseteq \mathbb{R}$	$\{X \in H\}$ is a finite union of sets $\{X = a_i\}$, hence in $\mathcal{G}$

Exercises

1. In the three-flip example with $\mathcal{F}_1 = \{\varnothing, A, A^c, \Omega\}$, is $Y = 2\,I_A + 5\,I_{A^c}$ $\mathcal{F}_1$-measurable? Write down $\{Y = 2\}$, $\{Y = 5\}$, and $\{Y \in [0, 3]\}$.
2. Let X be the number of heads. List $\{X = 0\}, \{X = 1\}, \{X = 2\}, \{X = 3\}$ and verify they partition Ω.
3. Find a σ-field $\mathcal{F}_3$ with $\mathcal{F}_1 \subsetneq \mathcal{F}_3 \subsetneq \mathcal{P}(\Omega)$ for which the number of heads is **not** measurable, and another for which it is.
4. Show that if $A \notin \mathcal{F}$ then I_A is not $\mathcal{F}$-measurable.
5. Let $X = \sum_{i=1}^{n} a_i I_{A_i}$ be as in the representation slide. Show $X^2 = \sum_{i=1}^{n} a_i^2 I_{A_i}$ and conclude X^2 is also simple.
6. Explain why a random variable taking countably infinitely many values (e.g. all positive integers) is not covered by the definition of "simple".

Outlook

Next lesson

The σ-field generated by a random variable. For X = number of heads in three flips, we found $\mathcal{F}_1$ too small and the power set possibly too large. Is there a smallest σ-field making X measurable?

Looking further ahead

Once random variables are in place, the tail σ-field and Kolmogorov's 0–1 law extend from sequences of events to sequences of random variables, with applications to convergence.